EGCP 4810 - Senior Design

Autonomous Golf Cart Project Proposal

By

The 2019-2020 Autonomous Vehicle Team:

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1 Definition Statement

The 2019-2020 Cedarville University Autonomous Vehicle team will build on the success of the 2018-2019 team by implementing and/or improving reversing, obstacle detection and avoidance, routing algorithms, and parking. We will accomplish these goals using LiDAR, GPS, and algorithmic technologies.

2 Abstract

The goal of our project is to build an autonomous golf cart that can facilitate short-distance driver-less transportation throughout the Cedarville University campus. This serves two immediate purposes:

1. Provide an alternate form of transportation around the campus.
2. Create an opportunity for Cedarville engineering students to gain hands-on experience with autonomous technology.

This golf cart will be able to intelligently route passengers as well as other cargo between buildings, parking lots, and intersections. The golf cart will accomplish autonomous navigation in a safe manner by detecting and avoiding obstacles. The cart will be able to autonomously park and reverse as well. We realize that the full scope of automating a vehicle will take longer than one year. Therefore, as we follow in the footsteps of last year's team, we will implement our changes so future senior design teams will have opportunities to improve our design and the functionality of the vehicle.

3 Background

This year's Autonomous Golf Cart project builds directly on the work of last year's team who accomplished the initial development phases. Because our project is inherited, it is critical to understand the current state of the autonomous golf cart so we can build upon it in an effective and relevant manner.

Last year's team's goals were to build a self-driving golf cart that could navigate itself from point to point over a short distance. They used GPS way-points to direct the golf cart from one place to another. An Arduino steered the cart using drive-by-wire. The team used a laptop to control the golf cart, but the operator could override commands from the laptop by manually using the brake or pushing an emergency stop button.

The Society of Automotive Engineers (SAE International) defines autonomy in cars in six levels *(Taxonomy and definitions for terms related to on-road motor vehicle automated driving systems (J3016 ground vehicle standard)—SAE Mobilus, 2014, Table 2)*. The autonomy achieved by last year's team can be classified as level three. This means the operator (the person sitting in the driver's seat) is not driving when the automated features are engaged, but may have to take control in certain situations. Therefore, the operator must be completely alert and engaged while the vehicle is in motion.

This year we will bring the golf cart to level four autonomy, meaning that the operator won't have to take control of the cart. The cart will operate autonomously, without operator intervention, in limited, defined conditions. These conditions are defined by the assumptions in section 4.1.

We are going to use LiDARs as our sensing platform as opposed to radars. This choice is due to budget constraints. LiDAR and radar pricing can range anywhere from $150-$75000. Since our team is limited by a modest four figure budget, enumerated in section 11, we are confined to budget LiDARs and radars. Available low-end radars have little documentation and support, which is integral to the development of our complete system. Therefore, we chose to research and pursue LiDAR technology.

There are three reasons that led to us deciding on the Livox Mid-40. The first was cost. This LiDAR was only $600 which, given its capabilities, is very cheap. The second reason is the extensive support and existing community surrounding this LiDAR. This includes documentation for using and converting the output of the LiDAR into a standard point-cloud data type. Finally, we chose this LiDAR because it is a 3D LiDAR instead of a 2D LiDAR, which can give us more information that we can use to help with navigation and obstacle avoidance.

4 Objectives

4.1 Assumptions

1. The vehicle will function on pedestrian walkways or low density roads.
2. The vehicle will operate in near-ideal mechanical and electrical conditions.

3. Potential obstacles in motion will not alter their course. In other words, if an object is moving along a particular path, the object will continue to move along that path regardless of the vehicle's presence.
4. All pre-programmed routes are traversable in the absence of obstacles.
5. All sensed objects are identifiable.
6. All relevant objects in the immediate path of the car can be sensed.
7. All non-sensed objects will not impact vehicle operation.
8. Stationary objects in the path of the car block the path.

4.2 Hard Requirements

4.2.1 Physical Requirements

1. The golf cart shall be able to transport the equivalent of four passengers.

4.2.2 Obstacle Avoidance Requirements

1. The obstacle avoidance system shall be able to react quickly enough to allow for safe handling of obstacles at all golf cart speeds.

4.2.3 Routing Requirements

1. The vehicle shall have the ability to autonomously deliver passengers to a specified area outside of heavy foot traffic, i.e. parking.
2. The vehicle shall have the ability to route to any location that is mapped to a destination selected from a predefined list. *Note:*
   (a) The user will choose a specific on-campus GPS way-point on the navigation computer. The campus will be sufficiently mapped so that all major academic and residential buildings (except Faith Hall) will be accessible.
   (b) "To route," as used above, means a static path chosen at the request time of the user. See section 4.3.2 for additional routing reach goals.
3. Moving objects in the path will trigger a stop.
   (a) If the moving object is not in the path after a certain interval of time the vehicle will continue its normal course.
   (b) If the moving object becomes stationary the vehicle remains stopped.

4.2.4 Vehicle Control Requirements

1. The vehicle shall have the ability to reverse through software.
2. The vehicle shall have the ability to move forward through software.
3. The vehicle shall have the ability to steer through software.
4. The vehicle shall have the ability to brake through software.
5. The vehicle shall be able to operate at the maximum speed of the cart.

4.3 Soft Requirements

4.3.1 Power Requirements

1. All electronic peripherals used on the golf cart shall be powered by the golf cart.
   (a) All sensors will be powered by the golf cart.
   (b) The control laptop will be powered by the golf cart.

4.3.2 Routing Requirements

1. The vehicle must navigate with the sole input of a destination from the user - meaning the vehicle chooses the best path dynamically to get from source to destination.
2. The vehicle will be able to determine if a previously moving obstacle has become a stationary obstacle.
3. Large stationary obstacles, which are blocking the path, will trigger a rerouting of the vehicle to optimize travel time.

4.3.3 Vehicle Display Requirements

1. The cart will use a web-based user-interface to display its real-time data.
2. The cart will have a mounted display for web-based user-interface in the vehicle.

4.4 Constraints

1. The vehicle be able to operate on pedestrian walkways (made of concrete, asphalt, or similar surfaces).
2. The vehicle must be able to make any round trip on campus without requiring external maintenance.

5 Strategy

The strategy for our project centers around the hardware and software teams and their different tasks (see section 8 for team breakdowns). Each team has different responsibilities and will develop independently until integration is necessary. Communication of the cart's software and hardware will occur through a software connection framework that our team will choose after sufficient research. This will allow the hardware team to gather data for obstacle avoidance and path navigation that can be easily used by the software team for developing obstacle handling algorithms. In addition, the software team will create navigation algorithms, which the hardware team can then test in different environments.

5.1 Hardware

1. Understand current hardware failures of the golf cart.
   (a) Run diagnostics on cart's hardware components using voltage probes.
   (b) Repair any severed connections and replace faulty components.
   (c) Test functionality of the system as a whole for lingering issues.
2. Complete reversing functionality.
   (a) Replace nylon and rope spool with more durable materials.
   (b) Determine functionality of existing braking detection system.
   (c) Investigate extent of software functionality for braking.
   (d) Modify and repair any necessary interfacing with existing braking software.
   (e) Test braking in both drive-by-wire and autonomous configurations.
3. Add LiDAR functionality.
   (a) Research and test ideal number of LiDAR sensors and different configurations.
   (b) Purchase and install LiDARs.
   (c) Test obstacle detection on the front and back sides of the vehicle.
   (d) Integrate LiDAR hardware with connection framework.
4. Add reversing functionality.
   (a) Test capacity of existing reverse functionality.
   (b) Design and build hardware for reversing.

   (c) Integrate reversing hardware with connection framework.

5. Weather proof hardware.
   (a) Add casing for wiring and hardware to protect against rain or wind damage.
   (b) 3D-print custom casing if necessary.

6. Modify golf cart's aesthetic for promotional purposes.
   (a) Ensure autonomous functionality is ready for promotional use (all necessary testing complete).
   (b) Paint cart with Cedarville colors and add university-related decals.
   (c) Add proper signage to satisfy safety requirements.
   (d) Generally clean or add covers to seats.

7. Add hardware for the real-time cart information display.

5.2 Software

1. Read and understand the previous year's code base.
   (a) Read and understand ConnectionHub and related components.
   (b) Read and understand GPS APIs and current integration with ConnectionHub.

2. Evaluate ConnectionHub and compare to existing open-source solutions.
   (a) Make a decision to proceed with current implementation or choose an existing framework.
   (b) If we move forward with a new framework, port existing integrations to new system.

3. Develop the routing system.
   (a) Develop a software routing service for traversing campus.
   (b) Collect GPS data points and assemble data into routes that can be used by the routing service.
   (c) Integrate routing service into the rest of the system.

4. Develop the object avoidance system.
   (a) Develop a LiDAR interface in order to integrate the hardware team's LiDAR unit into the rest of the software system.
   (b) Develop an object classification system and handling logic.

5. Develop a real-time cart information display.

6 Plan of Action

We will operate as two agile sub-teams, as outlined in section 8.1. Therefore, we chose Trello as our agile task tracking service. We will use a Gantt chart to present our high level initiative time-frames to our customers/advisers. A designated team member will align tasks (in agile jargon: epics - tasks, stories - sub-tasks) on Trello with the initiatives on the Gantt chart. Our preliminary Gantt chart, shown below, provides an overview of the estimated start date and duration of the major project initiatives. *Note: This timeline is not binding and our team will modify it as necessary.*

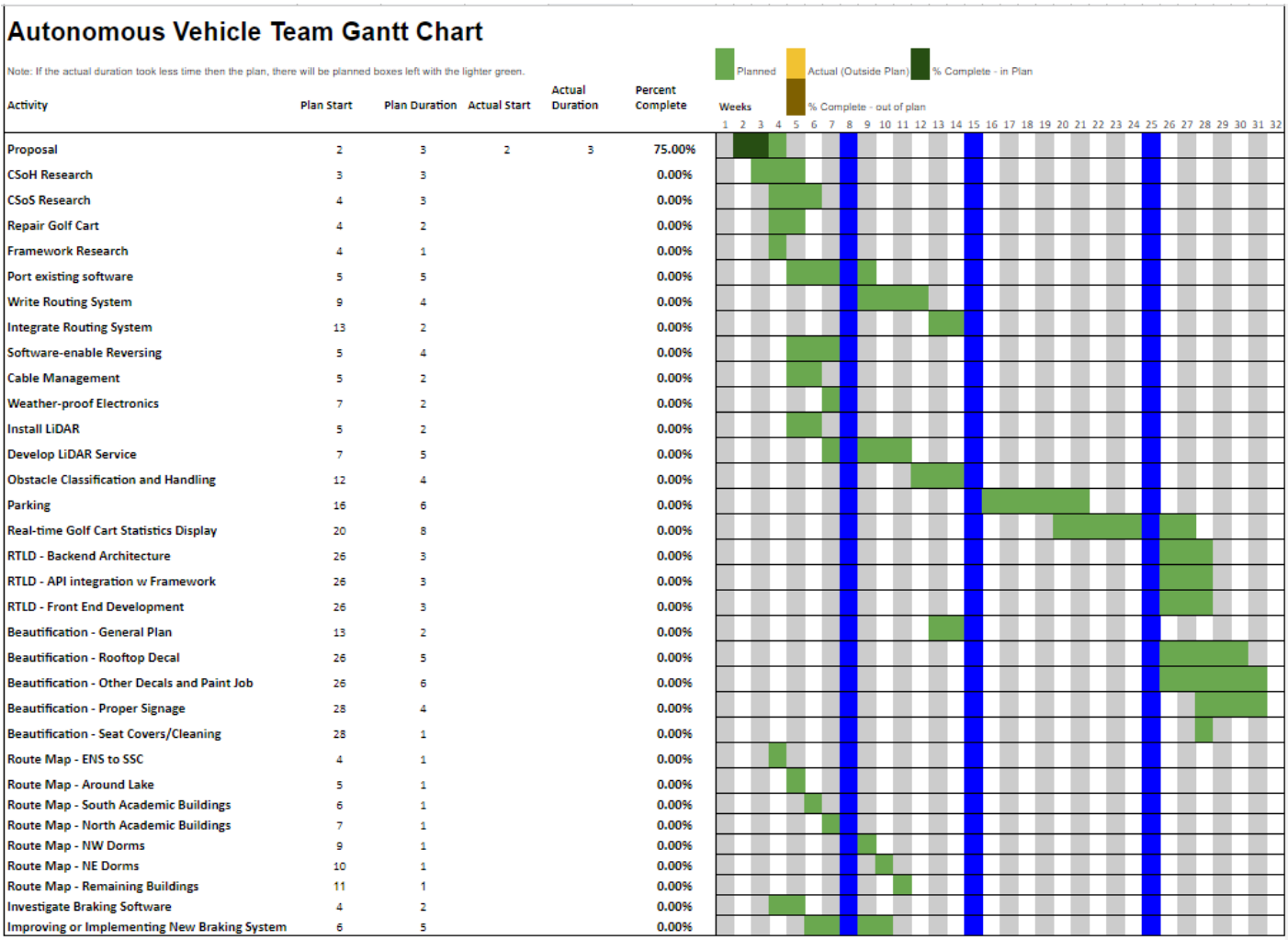

Figure 1: Epic Level Gantt Chart.

7 Milestones

- Milestone 1: Semester Week 9

  Software: By week 9, we aim to have chosen the connection framework. In addition, we aim to have mapped the route from the Engineering and Science Center to the Stephens Student Center.
  Hardware: We aim to have identified the location of the voltage drain and fixed the problem. We also aim to have cleaned up the wires that aren't straightforward and impede easy, user-friendly maintenance of the cart. We aim to return the golf cart to the state it was in at the end of Spring Semester 2019 - meaning the golf cart will complete 3 successful runs on a simple route.

- Milestone 2: Semester Week 16

  Software: By week 16, we aim to have mapped the academic buildings adjacent to the lake. We also aim to have ported the existing software to the framework we chose.
  Hardware: We aim to have physically installed the LiDAR systems on the golf cart and implemented a functional braking system.
  Beautification (Reach Goal): We aim to have decided on a plan for the beautification of the golf cart.

8 Division of Responsibilities

This year's team consists of six members, two electrical and four computer engineers.

8.1 Sub-teams

We created two sub-teams to handle specific needs of the vehicle:

- *Hardware Sub-Team*: Maintain and improve hardware and firmware technologies. The members are:
  - Emma Burgess
  - Daniel Garcia
  - Jake Lysack
- *Software Sub-Team*: Maintain and improve back-end API and connection framework code. The members are:
  - Anson Allard
  - Joel Beckmeyer
  - Daniel Parker

Since our team will operate in an Agile fashion, each sub-team member can work on the tickets they are interested in or are high priority. Therefore, this proposal will not enumerate each member's specific responsibilities.

8.2 Officer Designations and Responsibilities

In addition to the two sub-teams, we assigned two team members officer positions with the following responsibilities:

- *Chief Executive Officer (CEO)* - Anson Allard

  The CEO's responsibilities include, but are not limited to:

  1. Overseeing Agile development process for both sub-teams which includes:
     (a) Proposing timelines for tasks.
     (b) Working with sub-teams to set task priority.
  2. Being the point-person for customer and university relations.
  3. Handling administrative duties such as:
     (a) Submitting weekly reports.
     (b) Working with department faculty to enable the team's success.
     (c) Overseeing and orchestrating project proposals and presentations.

  Ultimately, the CEO is responsible to the customer for the team's performance. They are the first individual a customer will go to with questions, concerns, or complaints.

- *Chief Financial Officer (CFO)* - Jake Lysack

  The CFO's responsibilities include, but are not limited to:

  1. Creating and updating the yearly budget.
  2. Working with CEO and department faculty to ensure that the team obtains the necessary materials.

9 Reporting Plan

The team will submit weekly reports every Monday that will contain statements from the management, financial, software, and technical departments. These statements will provide updates on what occurred during the previous week, which would include any setbacks, accomplishments, and plans. Each individual will also provide their own update on what they personally accomplished during the previous week.

In addition to weekly reports, there will also be a weekly team meeting every Friday at 11 am. The focus of these meetings will be for each department and individual to inform the team of their progress, resolve any issues that occurred during the previous week, and create a plan for the upcoming week.

There will also be a Design Report that the team will complete by December 11 at noon. This will be a documentation of our progress and what we intend to develop further.

Lastly, the final report for the project will be completed in April, which will contain documentation of the full body of work from the year.

10 Testing Plan

10.1 Preliminary Tests

Our first category of tests will determine whether the cart is operating at the same level of functionality attained by last year's team. We will conduct three primary tests:

1. We will do a hardware check to ensure all components are working properly and the correct voltages are present.
2. We will conduct a drive-by-wire test to ensure navigation is possible through remote laptop controls.
3. We will do a GPS navigation test to make sure the full path navigation can be replicated.

10.2 New Tests

Our second category will test the ability of the vehicle as we build out features.

1. *GPS Routing* - We aim to dynamically route the golf cart around campus. Dynamically routing is defined as being able to navigate from any point to a pre-determined location regardless of the path and presence of obstacles. We intend to accomplish testing by generating multiple weighted graphs with varying weights between different nodes and using algorithms such as Dijkstra's Shortest Path First algorithm to determine the most efficient route. This test will determine how effectively the GPS and routing software can be used to complete this goal.
2. *Sensor Data Collection* - We will collect point-cloud data from all sensors mounted on the cart and manually verify that the data makes sense using the default visualization software that comes with the LiDARs. This test is designed to verify the integrity of the sensors as well as the granularity, accuracy, precision, and other related attributes of the data.
3. *Obstacle Sensing* - We will test the sensing system's ability to identify obstacles in both the forward and reverse directions by acquiring and categorizing objects in the system's output data. This test will involve iterating through new and diverse techniques (specifically from open-source platforms) that we intend to integrate into our final obstacle categorization system. Categorization groups include:
   (a) Categorizing the ground.
   (b) Categorizing moving obstacles.
   (c) Categorizing stationary obstacles.
4. *Reversing & Self-Parking* - This category will require running several tests to make sure the cart can properly reverse (whether during a route or in a parking area) as well as make sure the golf cart is able to properly park in a designated parking spot using GPS and LiDAR for assistance. The following are parking objectives:
   (a) Driving forward into a parking spot.
   (b) Backing into a parking spot.
   (c) Reverse to intersection when rerouting.

   This test is primarily visual in nature. In addition to observing how the cart behaves while parking, we will also record & assess sensor data to understand how the cart interprets reversing scenarios.
5. *Weather Resistance* - Finally, we will create scenarios that artificially alter the weather around the golf cart to make sure it can still fully function in inclement conditions without any incidents. This would include applying water to certain exposed parts of the cart to mimic rainy conditions.

11 Budget

Parts List and Costs			
Part	Description	Price ($)	Link
Advanced LiDAR (2)	Surveillance for main vehicle path	1500 (600 each + tax + shipping)	here
Basic LiDAR (4)	Surveillance for front and rear of vehicle	900 (200 each + tax + shipping)	here
Wireless Hotspot & Plan	Laptop requires internet usage	250	here
Jack & Stands	For general cart maintenance	175	n/a
Hardware for Braking System	Replacements/Additions to Existing Hardware	150	n/a
Laptop Power Supply	Hardware to use cart batteries to charge laptop	100	n/a
USB & Ethernet cables/hub	Communication between LiDAR & Laptop	75	n/a
Wire/Board Casing	Protect wire/boards from weather	50	n/a
Warning Posters	"Caution, Driverless Vehicle"	50	n/a
Paint[1]	Blue/Yellow for vehicle aesthetics	350	here
Stinger Decal[1]	For the roof of vehicle	200	n/a
Other Decals[1]	Sides/front/back of vehicle	200	n/a
Cleaning Supplies[1]	Seats/floor/body cleaning	50	n/a
	Total (Engineering Department)	3250	
	Total (Including External Funding)	4050	

[1] These items should be covered by funding outside of the Cedarville University Engineering Department (Approximately $800)